NeuroTechX

Building Japan's BMI/BCI Ecosystem

From research to global impact

Morgan Hough
Executive Director, NeuroTechX

Japan BMI/BCI Ecosystem Kick-off
Coral Capital, Tokyo / Hybrid
September 7, 2026

Make collaboration a repeatable process

Japan’s research strengths can become
shared pathways to real-world use.

01 / Connect

Bring users, researchers, builders, and partners into the same work.

02 / Build

Create tools, skills, and evidence that others can reuse.

03 / Translate

Give each promising project a next partner and a next milestone.

NeuroTechX connects people to the field

Community

Local chapters and a global network create places to meet and collaborate.

Education

Accessible learning resources help people enter neurotechnology and develop skills.

Professional development

Projects and connections help talent find its next opportunity.

A community can make the next collaboration easier to start.

Source: NeuroTechX: mission and three pillars

Start with the strengths already in the room

Discovery and engineering

Brain science
Neural engineering
AI and robotics

Translation and production

Rehabilitation
Medical devices
Precision manufacturing

Organize around a user need that requires several of these strengths.

Context: strengths identified in the organizers’ event brief.

An ecosystem needs working handoffs

User need
Define value

→

Research
Test a mechanism

→

Product
Build reliability

→

Use
Support adoption

Feedback from users and real operating conditions

For every handoff: an owner, evidence, and a receiving partner.

Different interfaces need different partners

For an implanted BCI project

Which clinical team and intended users shape the design?

Who supports implantation, follow-up, and long-term use?

For a non-invasive BCI project

Where and by whom will it be used?

How will setup, signal quality, comfort, and repeat use be evaluated?

Choose a use case before choosing a collaboration model.

At the AI interface, agree on the test

1. What does the model add over a simple baseline?
2. Does the result hold across sessions and people?
3. What happens when the model is uncertain or wrong?
4. Does the complete system help the intended user?

A shared evaluation plan makes collaboration concrete.

A working example: shared BCI benchmarks

MOABB Mother of All BCI Benchmarks

What exists

An open-source framework for comparing BCI algorithms on public EEG datasets.

A possible Japan–global project

Reproduce a baseline together, document the protocol, and contribute a useful improvement.

Give a new partnership something small and verifiable to build.

Source: MOABB: project description and evaluation example

Give people a path from interest to contribution

Enter

A welcoming introductory session.

Explain the problem and prerequisites.

Practice

A guided workshop using a reproducible example.

Pair different disciplines.

Contribute

A scoped issue and an available mentor.

Publish what the team learned.

Measure whether people return and finish useful work.

Make international collaboration specific

Start with one shared artifact

A benchmark replication
A bilingual technical guide
A jointly reviewed project brief

Give it a working agreement

One owner on each side
A regular meeting time
A review date and success criterion

Translate the work so that the next team can participate.

Fund the work between the milestones

Work that needs an owner

Data and protocol documentation
Tool maintenance and testing
Mentoring and coordination
User feedback and follow-through

A useful partner offer

An engineer’s time
A mentor or domain specialist
A project venue or equipment
Support for a defined deliverable

Ask partners to support a deliverable with a named steward.

A proposed first 90 days

Days 1–30 / Map

Identify willing participants.

Choose one problem and two project leads.

Days 31–60 / Build

Run a joint workshop.

Produce a reproducible first artifact.

Days 61–90 / Review

Demonstrate the work.

Publish lessons and decide the next step.

Leave today with an owner and a date for the next conversation.

Measure what the collaboration makes possible

Question	Evidence to bring to the review
Did people become contributors?	Completed tasks and returning participants.
Can another team use the work?	An independent reproduction or documented reuse.
Did the project move forward?	A receiving partner and an agreed next milestone.
Can the effort continue?	A maintainer, time allocation, and next review date.

What could each of us bring?

Researchers and clinicians

A meaningful question, a protocol, or a domain mentor.

Startups and industry

A concrete engineering problem and someone available to work on it.

Builders and students

A reproducible example, a documented tool, or time to contribute.

Funders and ecosystem organizers

Support for coordination, a scoped deliverable, and a review.

One problem. Two partners. One next step.

What could we build together

in the next 90 days?

neurotechx.org/community

github.com/NeuroTechX/moabb

Sources and scope

Organization and visual reference
NeuroTechX: about and mission
NeuroTechX: website visual identity

Open-source example
MOABB: official repository and documentation

Event context
Organizer-provided event brief; September 7, 2026, Tokyo.

The collaboration framework and 90-day plan are proposals.
This draft does not announce organizational commitments. Sources checked September 4, 2026.